



Chapter 2

Discrete-Time Signals in the Time Domain



§ 2 Discrete-Time Signals In the Time-Domain



§ 2.1 Time-Domain Representation

- Discrete-time signals $x[n]$ --represented as *sequences* with *argument* n being an integer in the range $-\infty < n < \infty$
- The *nth* sample value of sequence $x[n]$ is also denoted as $x[n]$
- values of *argument* n --defined only for integer and undefined for noninteger values of n
- Discrete-time signal represented by $\{x[n]\}$



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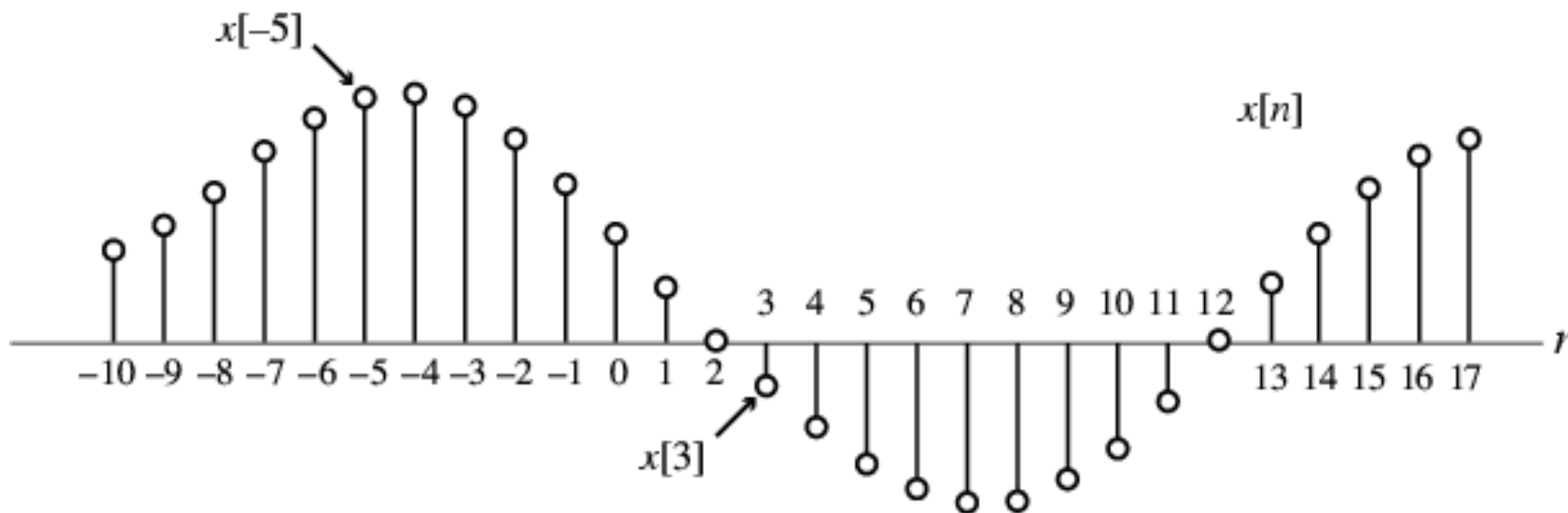
- Discrete-time signal may also be written as a sequence of numbers inside braces:
$$\{x[n]\} = \{\dots, -0.2, 2.2, 1.1, 0.2, -3.7, 2.9, \dots\}$$

↑
- The arrow is placed under the sample at time index $n = 0$
- In the above, $x[-1] = -0.2$, $x[0] = 2.2$, $x[1] = 1.1$, etc.

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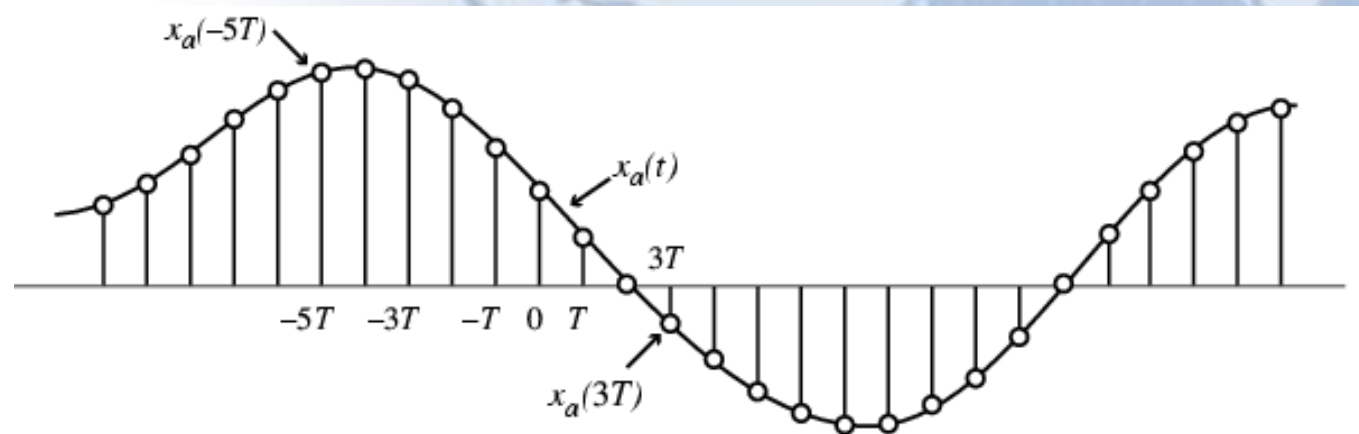
- *Graphical* representation of a discrete-time signal with real-valued samples is as shown below:



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In some applications, a discrete-time sequence $\{x[n]\}$ may be generated by periodically sampling a continuous-time signal $x_a(t)$ at uniform intervals of time



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- Here, n -th sample is given by
$$x[n] = x_a(t) |_{t=nT} = x_a(nT), \quad n = \dots, -2, -1, 0, 1, \dots$$
- The spacing T between two consecutive samples is called the *sampling interval* or *sampling period*
- Reciprocal of sampling interval T , denoted as F_T , is called the *sampling frequency*:

$$F_T = 1/T$$

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- Unit of sampling frequency is cycles per second, or hertz (Hz), if T is in seconds
- Whether or not the sequence $\{x[n]\}$ has been obtained by sampling, the quantity $x[n]$ is called the n -th sample of the sequence
- $\{x[n]\}$ is a *real sequence*, if the n -th sample $x[n]$ is real for all values of n
- Otherwise, $\{x[n]\}$ is a *complex sequence*

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- A complex sequence $\{x[n]\}$ can be written as

$$\{x[n]\} = \{x_{\text{re}}[n]\} + j\{x_{\text{im}}[n]\} \quad (2.4)$$

where $x_{\text{re}}[n]$ and $x_{\text{im}}[n]$ are the real and imaginary parts of $x[n]$

- The complex conjugate sequence of $\{x[n]\}$ is given by

$$\{x^*[n]\} = \{x_{\text{re}}[n]\} - j\{x_{\text{im}}[n]\}$$

- Often the braces are ignored to denote a sequence if there is no ambiguity

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Examples:

- $\{x[n]\} = \{\cos 0.25n\}$ is a real sequence
- $\{y[n]\} = \{e^{j0.3n}\}$ is a complex sequence
- We can write

$$\begin{aligned}\{y[n]\} &= \{\cos 0.3n + j\sin 0.3n\} \\ &= \{\cos 0.3n\} + j\{\sin 0.3n\} \\ \text{where } \{y_{\text{re}}[n]\} &= \{\cos 0.3n\} \\ \{y_{\text{im}}[n]\} &= \{\sin 0.3n\}\end{aligned}$$

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2.1.1 Length of a discrete-time signal

- A discrete-time signal may be a finite-length or an infinite-length sequence
- *Finite-length* (also called finite-duration or finite-extent) sequence is defined only for a finite time interval: $N_1 \leq n \leq N_2$ where $-\infty < N_1$ and $N_2 < \infty$ with $N_1 \leq N_2$
- Length or duration of the above finite-length sequence is

$$N = N_2 - N_1 + 1$$

Zero-padding—append with zero-valued samples.

Vector form $\mathbf{x}[n] = [x[0] \ x[1] \ \dots \ x[N-1]]^t$

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- A right-sided sequence $x[n]$ has zero-valued samples for $n < N_1$



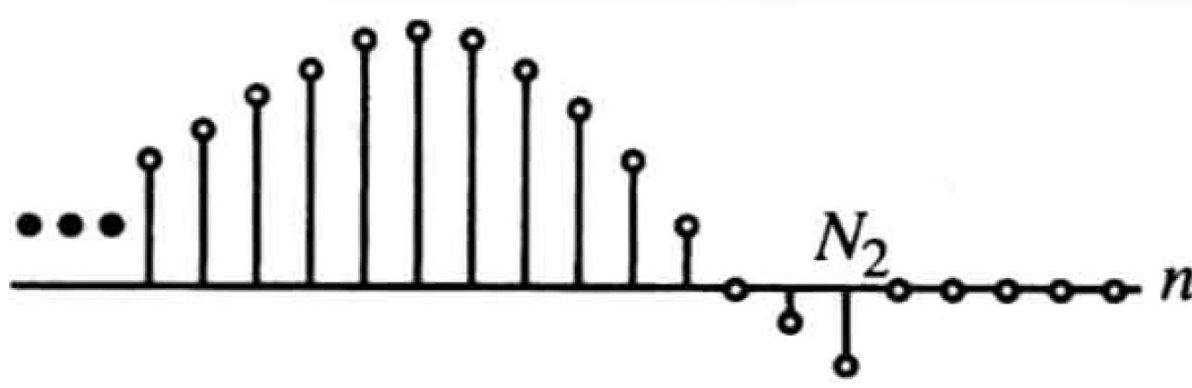
A right-sided sequence

- If $N_1 \geq 0$, a right-sided sequence is called a **causal sequence**.

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- A left-sided sequence $x[n]$ has zero-valued samples for $n > N_2$



A left-sided sequence

- If $N_2 \leq 0$, a left-sided sequence is called a **anticausal sequence**.

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2.1.2 Strength of a Discrete-Time Signal

The strength of a DT Signal usually is given by its norm.

L_p - norm of $\{x[n]\}$ is defined by

$$\|x\|_p = \left(\sum_{n=-\infty}^{\infty} |x[n]|^p \right)^{1/p} \quad (2.9)$$

p is a positive integer.

In practice, the value of p used is typically 1 or 2 or ∞ .

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Peak absolute value

L_∞ - norm

$$\|x\|_\infty = |x[n]|_{\max} \quad (2.10)$$

For a length-N sequence,

$\|x\|_2 / \sqrt{N}$ is the **root-mean-squared(rms)** value of $\{x[n]\}$, and

$\|x\|_1 / \sqrt{N}$ is the **mean absolute value** of $\{x[n]\}$.

It can be shown that $\|x\|_2 \leq \|x\|_1$

M-file in MATLAB [norm\(x,1\)](#) [norm\(x,2\)](#) [norm\(x,inf\)](#)

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- One application of the norm is in estimating the error in approximation of a DT signal by another DT signal **in some sense**.

For example, a length- N sequence $y[n]$, and another $x[n]$, $0 \leq n \leq N-1$

mean-squared error(MSE)

$$MSE = \frac{1}{N} \left(\|y[n] - x[n]\|_2 \right)^2 \quad (2.11)$$

relative error

$$E_{rel} = \frac{\|y[n] - x[n]\|_2}{\|x[n]\|_2} \quad (2.12)$$

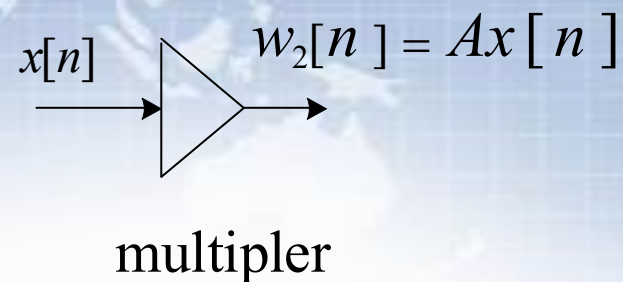
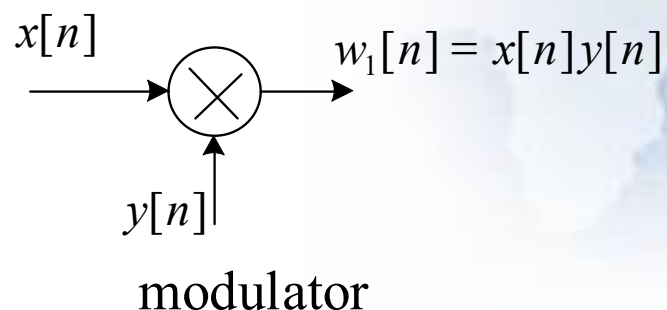
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§ 2.2 The Basic Operations On Sequences

2.2.1 Elementary Operations

modulator(product or windowing)

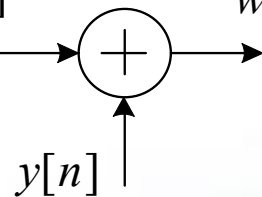


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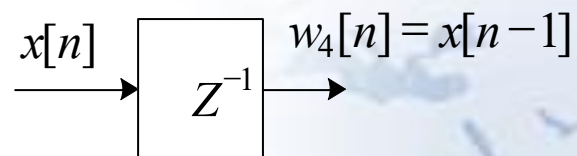


$$x[n] \rightarrow \text{adder} \rightarrow w_3[n] = x[n] + y[n]$$



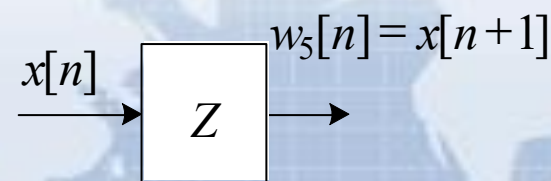
addition(subtraction)

adder

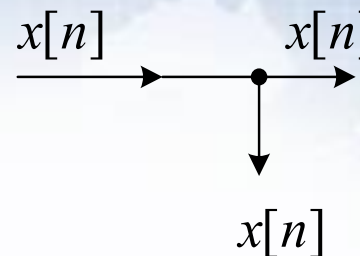


unit delay

pick-off node



unit advance



time-reversal (folding operation)

$x[-n]$

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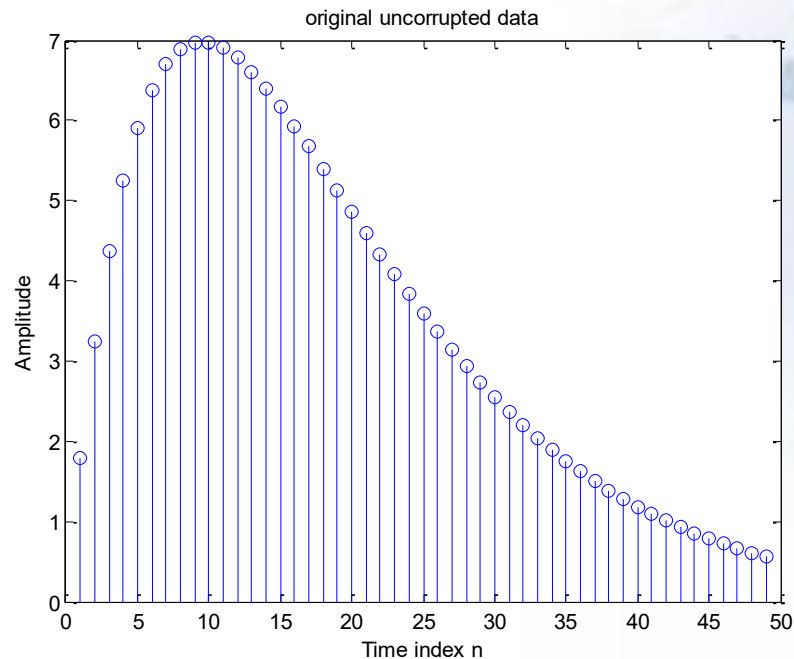
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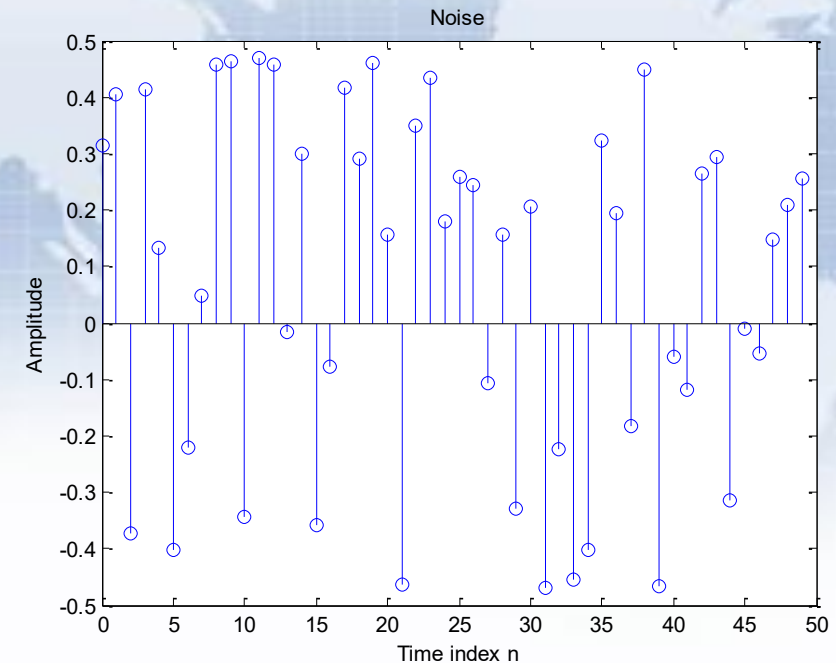
Example 2.1 Ensemble average

$$s[n] = 2[n(0.9)^n]$$

original uncorrupted data



Noise



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Vector form

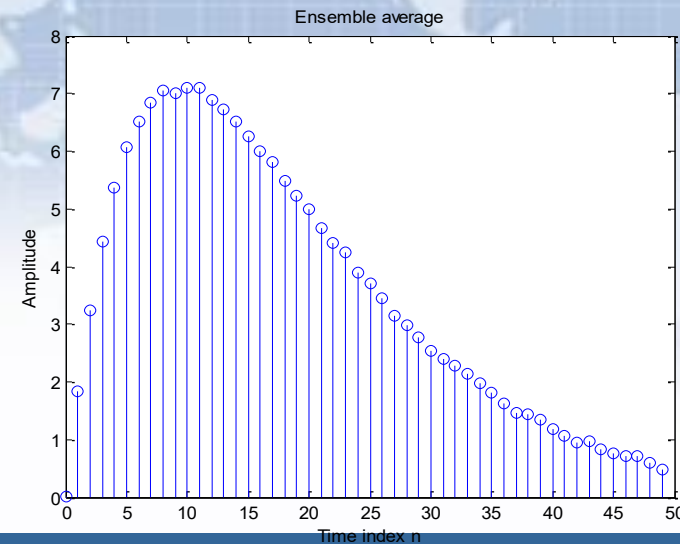
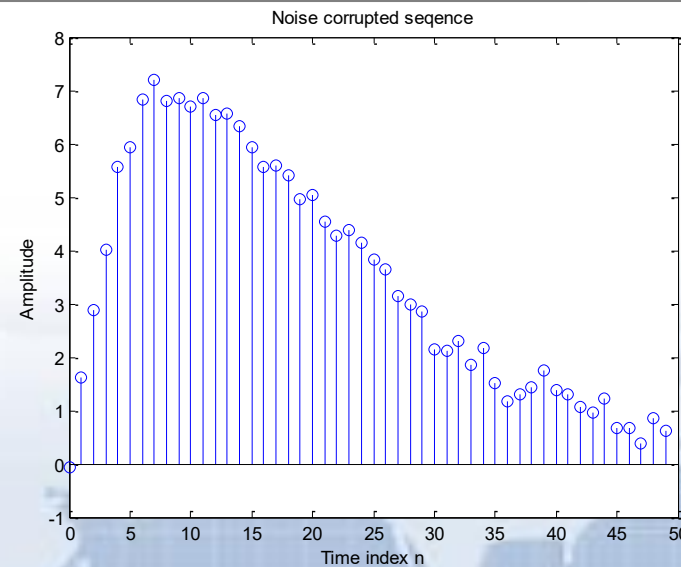
$$\mathbf{s} = [s[0] \ s[1] \ \dots \ s[N-1]]$$

The i -th measurement of data vector

$$\mathbf{x}_i = \mathbf{s} + \mathbf{d}_i$$

Ensemble average

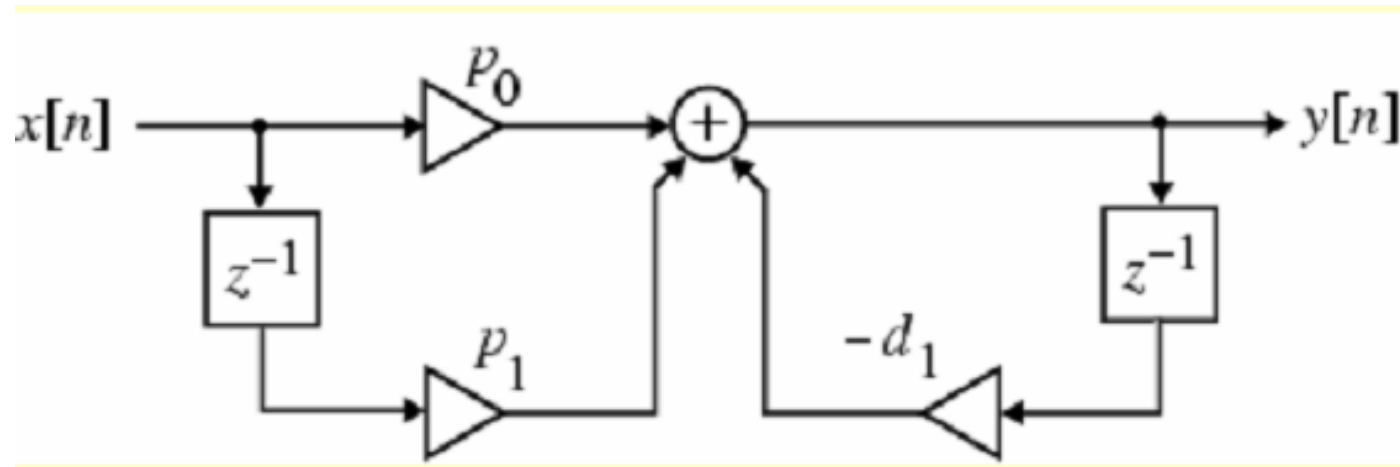
$$\begin{aligned} \mathbf{x}_{ave} &= \frac{1}{K} \sum_{i=1}^K (\mathbf{x}_i) = \frac{1}{K} \sum_{i=1}^K (\mathbf{s} + \mathbf{d}_i) \\ &= \mathbf{s} + \frac{1}{K} \left(\sum_{i=1}^K \mathbf{d}_i \right) \end{aligned}$$



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2.2.2 Combination of Elementary Operations



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2.2.3 Convolution Sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

is called the **convolution sum** of the sequences $x[n]$ and $h[n]$ and represented compactly as

$$y[n] = x[n] \odot h[n]$$

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- **Example** Develop the sequence $y[n]$ generated by the convolution of the sequences $x[n]$ and $h[n]$:

$$x[n] = h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

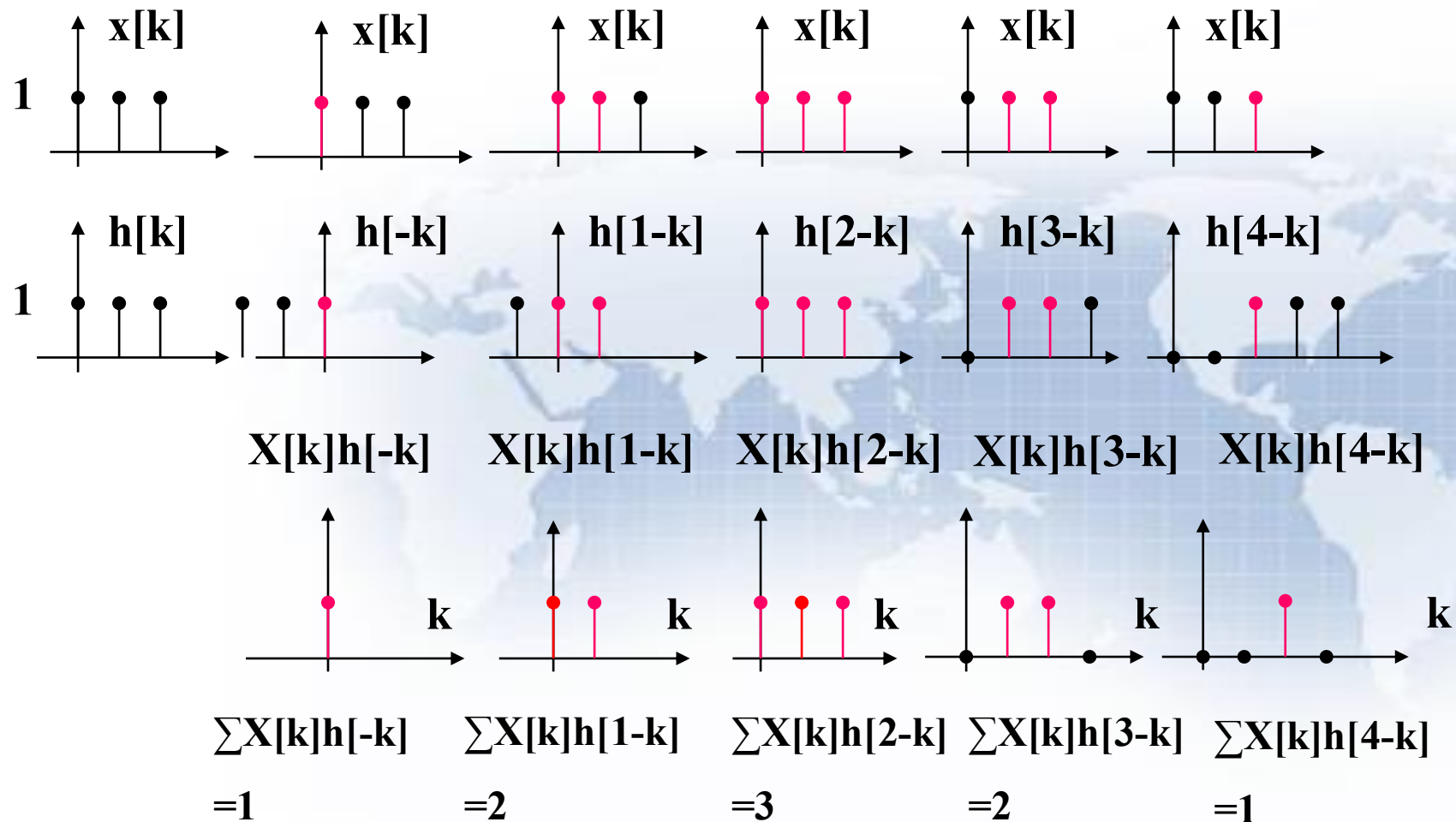
$$y[n] = x[n] \circledast h[n] = \sum_{k=0}^{\infty} x[k]h[n-k]$$

$$y[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 2\delta[n-3] + \delta[n-4]$$

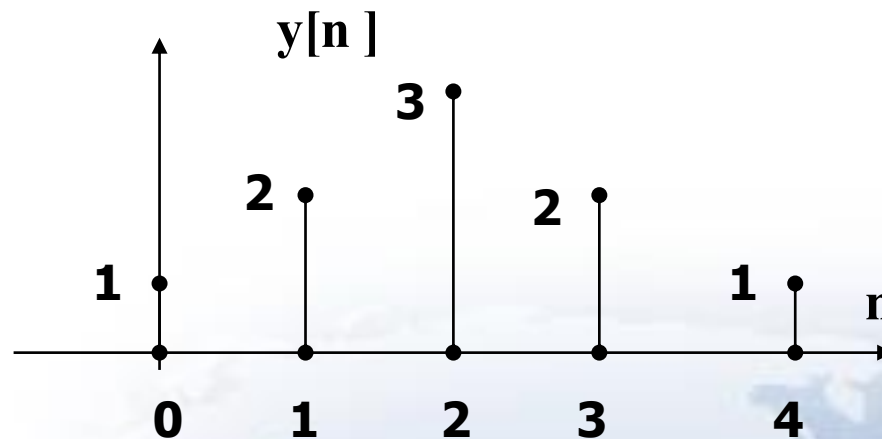
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$$x[n] = h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$



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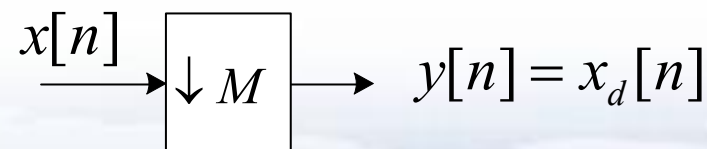
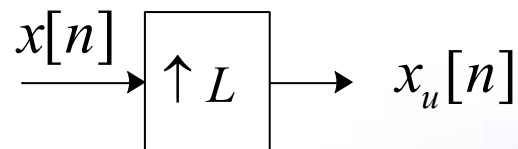


- In general, if the lengths of the two sequences being convolved are M and N , then the sequence generated by the convolution sum is of length $M+N-1$
- **Read** example 2.5 and 2.6 **by yourself**.

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2.2.4 Sampling rate alteration



Up-sampling

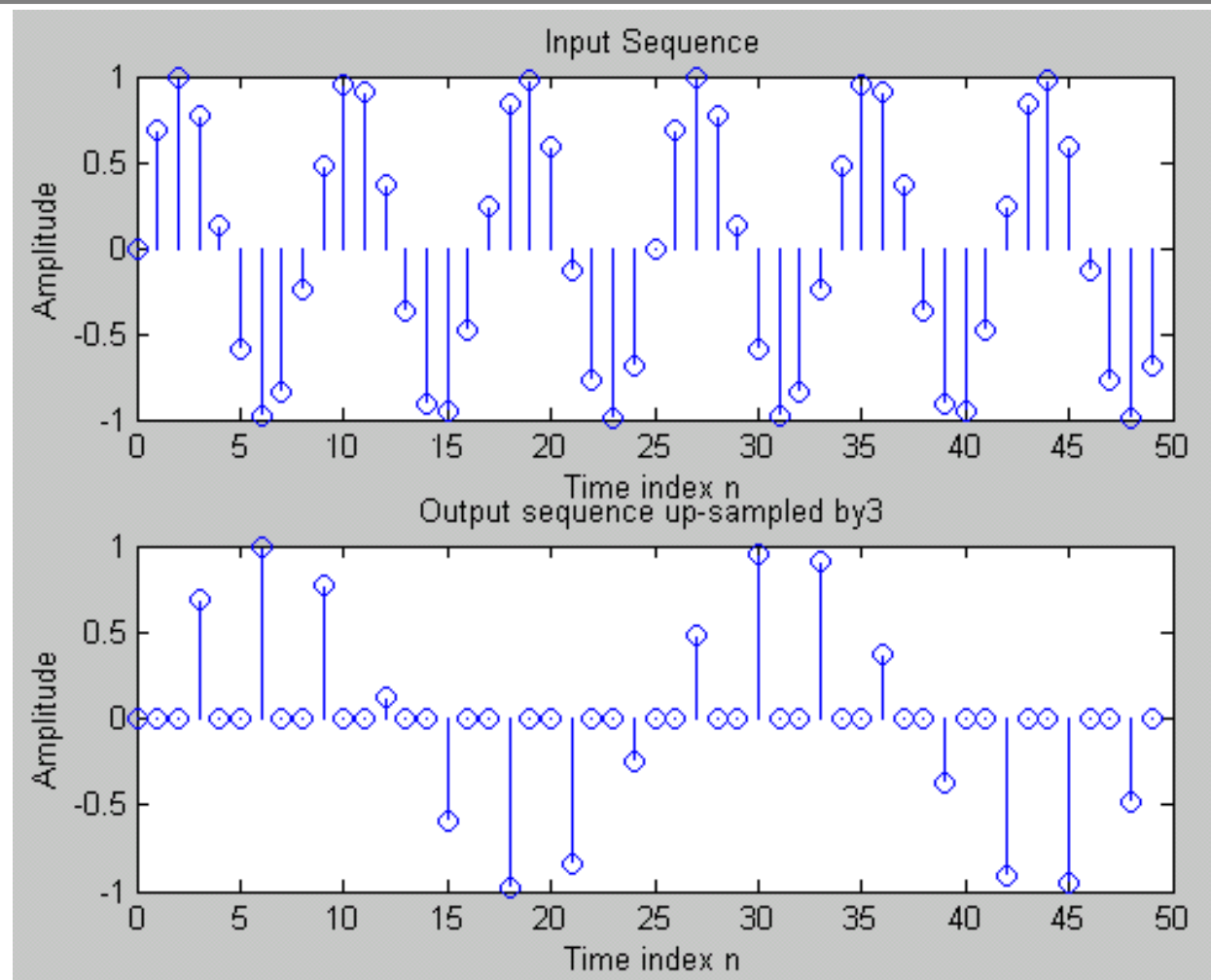
$$x_u[n] = \begin{cases} x[n/L], n = 0, L, \pm 2L, \dots \\ 0, otherwise \end{cases}$$

Down-sampling

$$x_d[n] = x[Mn]$$

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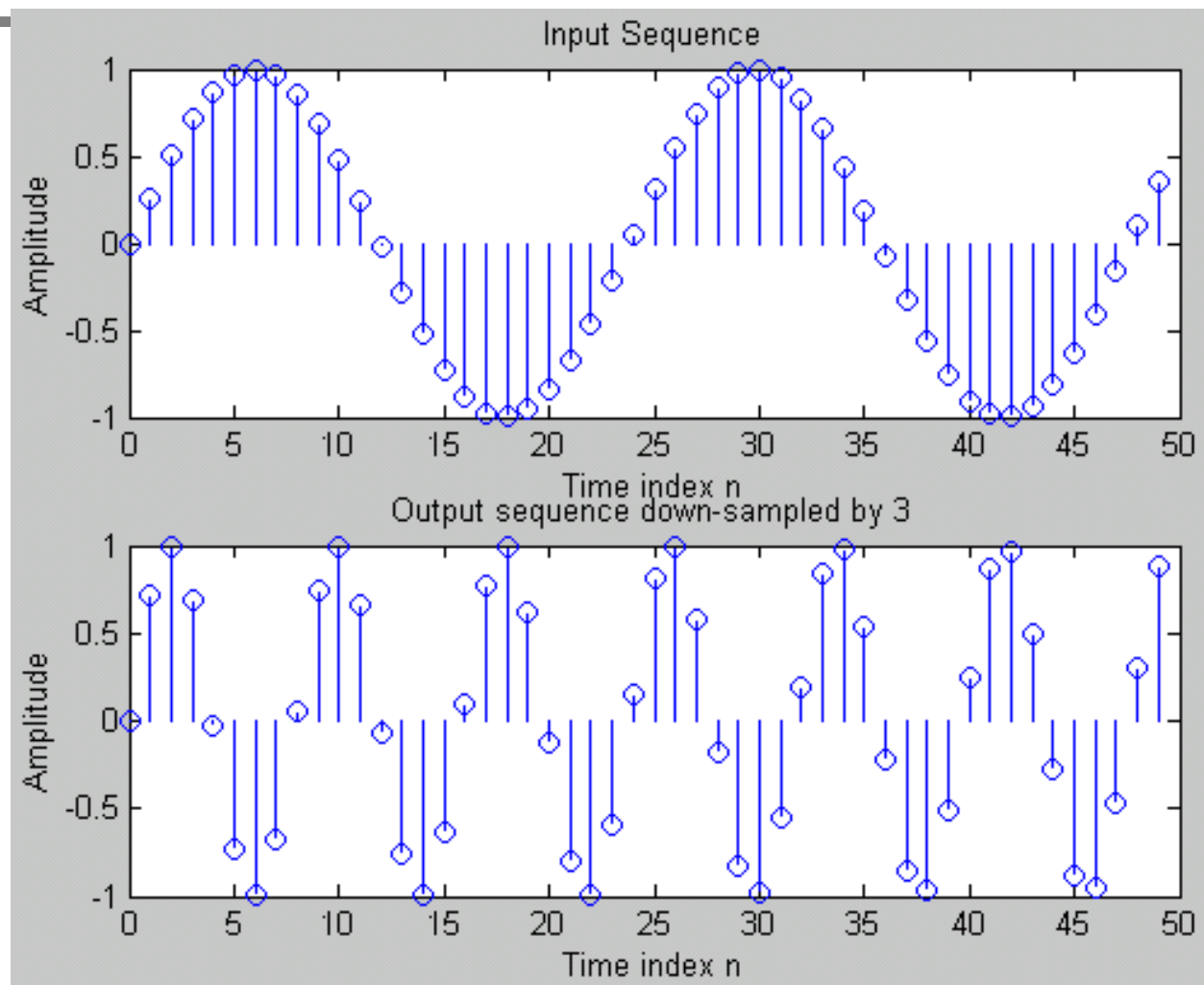
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Up-sampling (L=3)

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Down-sampling ($M=3$)

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§ 2.3 Operations on Finite-length Sequences

- Some of the operations on sequences described in previous section are **not applicable** to finite-length sequences.
- Consider length- N sequences $x[n]$ defined for $0 \leq n \leq N-1$. Sample values of such sequences are equal to zero for values of $n < 0$ and $n \geq N$.
- $x[-n]$ is defined for the range $-(N-1) \leq n \leq 0$.

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- For any arbitrary integer n_0 , $x_1[n] = x[n - n_0]$ is no longer defined for the range $0 \leq n \leq N-1$
- We thus need to define another type of operation that will always keep the operated sequence defined for the range $0 \leq n \leq N-1$.

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2.3.1 circular time-reversal operation

Modulo operation $\langle m \rangle_N = m \text{ modulo } N$

Integer m $r = m + lN$

Integer r: [0, N-1]

$$r = \langle m \rangle_N$$

residue

Example $N = 7, m = 25, r = 25 + 7l = 25 - 7 \times 3 = 4$

Thus, $\langle 25 \rangle_7 = 4$

$$N = 7, m = -15, r = -15 + 7l = -15 + 7 \times 3 = 6$$

Thus, $\langle -15 \rangle_7 = 6$

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circular time-reversal operation

$$\{y[n]\} = \{x[\langle -n \rangle_N]\}$$

defined for $0 \leq n \leq N-1$

$$x[\langle -n \rangle_N] = \begin{cases} x[N-n], & \text{for } 1 \leq n \leq N-1 \\ x[n], & \text{for } n = 0 \end{cases}$$

$$\text{Q } \langle -n \rangle_N = \langle N-n \rangle_N \quad \text{for } 1 \leq n \leq N-1$$

$$\therefore x[\langle -n \rangle_N] = x[\langle N-n \rangle_N]$$

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2.3.2 Circular Shift of a Sequence

- The **circular shift**, is defined using a modulo operation:

$$x_c[n] = x[\langle n - n_o \rangle_N]$$

- For $n_o > 0$ (right circular shift), the above equation implies

$$x_c[n] = \begin{cases} x[n - n_o], & \text{for } n_o \leq n \leq N - 1 \\ x[N - n_o + n], & \text{for } 0 \leq n < n_o \end{cases}$$

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2.3.3 Classification of Sequences

1. Classification based on Symmetry

complex sequence $x[n]$

Conjugate-symmetry sequence

$$x^*[n] = x[-n]$$

Conjugate-antisymmetry sequence

$$x^*[n] = -x[-n]$$

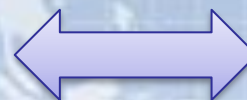
real $x[n]$

Even

$$x[n] = x[-n]$$

Odd

$$x[n] = -x[-n]$$



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$x[n]$ is complex sequence,

Conjugate-symmetry part

$$x_{cs}[n] = \frac{1}{2} \{x[n] + x^*[-n]\}$$

Conjugate-antisymmetry part

$$x_{ca}[n] = \frac{1}{2} \{x[n] - x^*[-n]\}$$

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Example 2.8 A length-7 sequence($n: [-3, 3]$):

$$\{g[n]\} = \{0, \quad 1+j4, \quad -2+j3, \quad 4-j2, \quad -5-j6, \quad -j2, \quad 3\}$$



$$\{g^*[n]\} = \{0, \quad 1-j4, \quad -2-j3, \quad 4+j2, \quad -5+j6, \quad j2, \quad 3\}$$



$$\{g^*[-n]\} = \{3, \quad j2, \quad -5+j6, \quad 4+j2, \quad -2-j3, \quad 1-j4, \quad 0\}$$



$$\{g_{cs}[n]\} = \{1.5, 0.5+j3, -3.5+j4.5, \quad 4, \quad -3.5-j4.5, 0.5-j3, 1.5\}$$



$$\{g_{ca}[-n]\} = \{-1.5, 0.5+j, 1.5-j1.5, \quad -j2, \quad -1.5-j1.5, -0.5+j, 1.5\}$$



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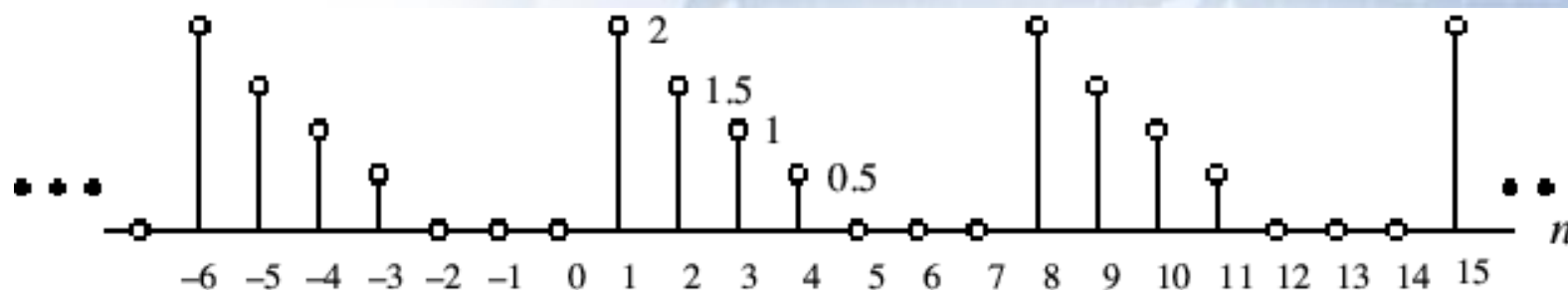
2. Classification of Sequences based on periodicity

- A sequence satisfying the periodicity condition

$$\tilde{x}[n] = \tilde{x}[n + kN] \quad \text{for all } n$$

$\tilde{x}[n]$ is called an **periodic sequence**

• Example



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fundamental periods $N_f = \text{smallest } N$

Adding two periodic sequences with different fundamental periods

$$\tilde{y}[n] = \tilde{x}_a[n] + \tilde{x}_b[n] \quad N_a \quad N_b$$

The fundamental period:

$$N = LCM(N_a, N_b) = \frac{N_a N_b}{GCD(N_a, N_b)}$$

LCM()---*least common multiple*

GCD()---*greatest common divider*

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Example
$$f[n] = \cos \frac{\pi}{8} n + \frac{1}{2} \sin \frac{\pi}{6} n$$

Its fundamental period is 48, because the GCD of 16 and 12 is 4.

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3. Energy and Power Signals

- Total energy of a sequence $x[n]$ is defined by

$$\mathcal{E}_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- The average power of a sequence is defined by

$$P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2$$

Energy signal: a finite energy signal with zero average power.

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- Example - **Power signal**: an infinite energy signal with finite average power.

$$x[n] = \begin{cases} 3(-1)^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

- **Note:** $x[n]$ has infinite energy
- Its average power is given by

$$P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \left(9 \sum_{n=0}^K 1 \right) = \lim_{K \rightarrow \infty} \frac{9(K+1)}{2K+1} = 4.5$$



4. Other Types of Classification

- A sequence $x[n]$ is said to be **bounded** if

$$|x[n]| \leq B_x < \infty$$

- **Example** - The sequence $x[n]=\cos(0.3\pi n)$ is a bounded sequence as

$$|x[n]| = |\cos(0.3\pi n)| \leq 1$$

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- A sequence $x[n]$ is said to be **absolutely summable** if

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

- **Example** - The sequence $y[n] = \begin{cases} 0.3^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$

is an absolutely summable sequence as

$$\sum_{n=0}^{\infty} |0.3^n| = \frac{1}{1-0.3} = 1.42857 < \infty$$

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- A sequence $x[n]$ is said to be **square-summable** if

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

- **Example** - The sequence $h[n] = \frac{\sin 0.4n}{\pi n}$

is square-summable but not absolutely summable

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- Example - The sequence $\tilde{y}[n]$ obtained by adding an absolutely summable sequence $x[n]$ with its replicas shifted by integer multiples of N ,

$$\tilde{y}[n] = \sum_{k=-\infty}^{\infty} x[n + kN]$$

$\tilde{y}[n]$ is called an N -periodic extension of $x[n]$.

It is **neither** absolutely summable **nor** square-summable .

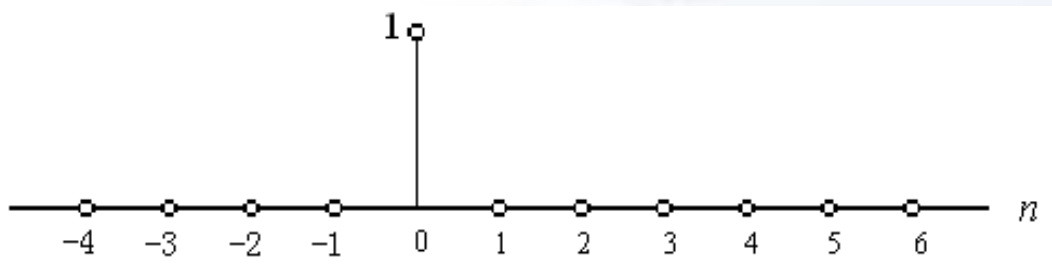
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§ 2.4 Typical Sequences and Representation

2.4.1 Basic Sequences

- Unit sample sequence



$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

- Unit step sequence



$$\mu[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

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2.4.3 Representation of an Arbitrary Sequence by impulses

- An arbitrary sequence can be represented in the time-domain as a weighted sum of some basic sequence and its delayed (advanced) versions

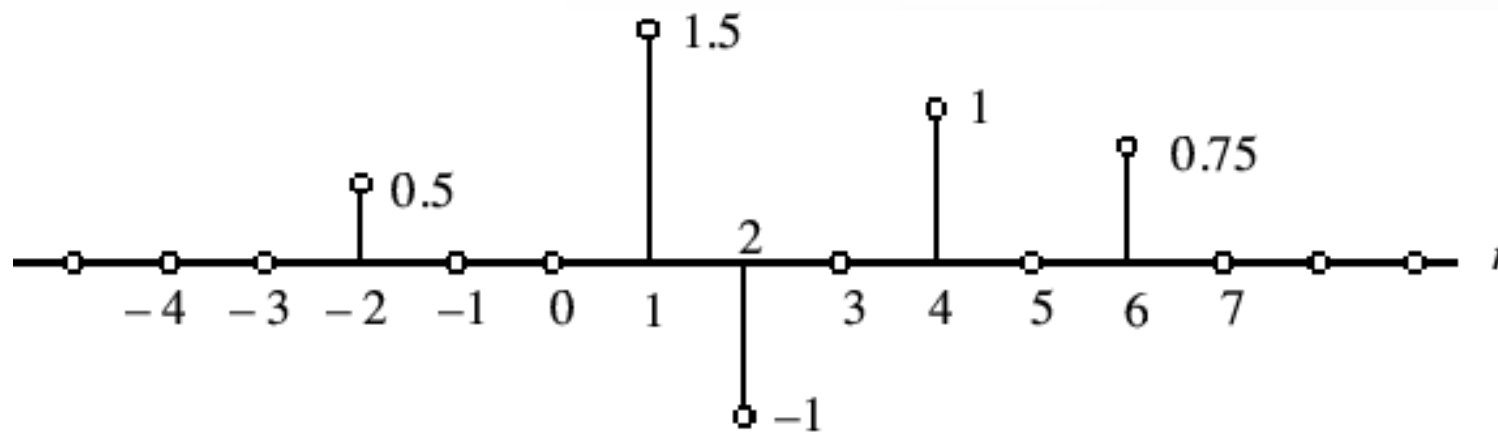
$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

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- **Example**



$$x[n] = 0.5\delta[n+2] + 1.5\delta[n-1] - \delta[n-2] + \delta[n-4] + 0.75\delta[n-6]$$

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Rectangular Window Sequence

- **Box-car sequence**

$$w_R[n] = \begin{cases} 0, & n < N_1, \\ 1, & N_1 \leq n \leq N_2 \\ 0, & n > N_2 \end{cases} \quad (2.56)$$

- **Windowing**

$$x[n] \cdot w_R[n] = \begin{cases} 0, & n < N_1, \\ x[n], & N_1 \leq n \leq N_2, \\ 0, & n > N_2, \end{cases} \quad (2.57)$$

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- Sinusoidal Sequence

$$x[n] = A(\cos \omega_0 n + \phi), -\infty < n < \infty$$

where A is the *amplitude*, ω_0 is the *normalized angular frequency*, and ϕ is the *phase* of $x[n]$.

- The unit of ω_0 and ϕ is **radians/sample**

- Often in practice, ω_0 is expressed as

$$\omega_0 = 2\pi f_0$$

Where f_0 is *normalized frequency* in **cycles/sample**.

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- Sinusoidal Sequence

$$x[n] = A(\cos \omega_0 n + \phi), -\infty < n < \infty$$

– We can write as

$$x[n] = x_i[n] + x_q[n]$$

$x_i[n]$ is the *in-phase components* of $x[n]$ and

$x_q[n]$ is the *quadrature components* of $x[n]$.

$$x_i[n] = A \cos \phi \cos(\omega_0 n) \quad x_q[n] = -A \sin \phi \sin(\omega_0 n)$$

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- Exponential Sequence

$$x[n] = A\alpha^n, -\infty < n < \infty \quad (2.51)$$

when A and α are **real or complex** numbers.
By expressing

$$\alpha = e^{(\sigma_0 + j\omega_0)}, A = |A|e^{j\phi}$$

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The Complex Exponential Sequence

we can rewrite

$$x[n] = |A|e^{\sigma_0 n} e^{j(\omega_0 n + \phi)} \quad (2.52a)$$

$$= |A|e^{\sigma_0 n} \cos(\omega_0 n + \phi) + j|A|e^{\sigma_0 n} \sin(\omega_0 n + \phi) \quad (2.52b)$$

And $x[n] = x_{re}[n] + jx_{im}[n]$

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$$x_{re}[n] = |A|e^{\sigma_0 n} \cos(\omega_0 n + \phi)$$

$$x_{im}[n] = |A|e^{\sigma_0 n} \sin(\omega_0 n + \phi)$$

The complex sinusoidal sequence

$$e^{j\omega_0 n} = \cos \omega_0 n + j \sin \omega_0 n$$

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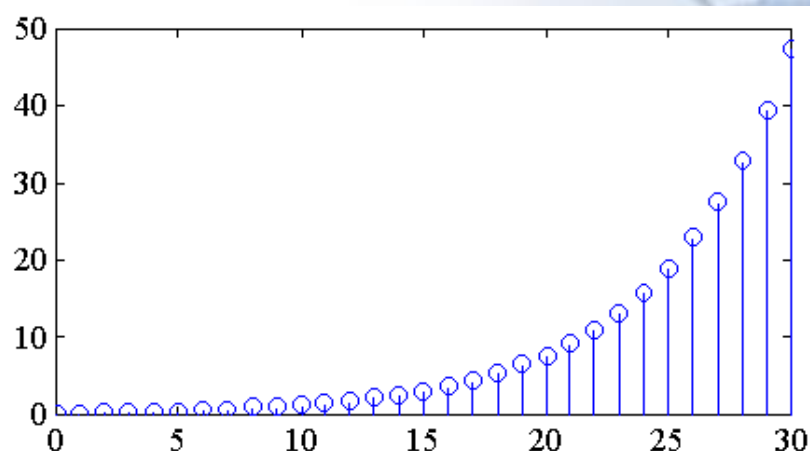


Real Exponential Sequence

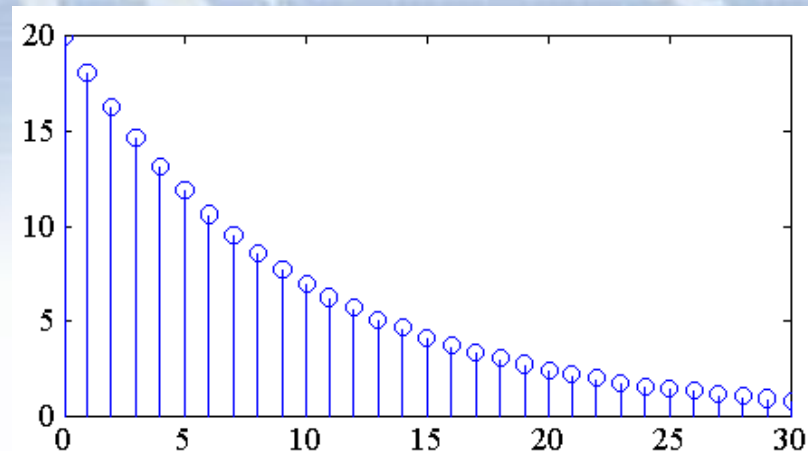
$$x[n] = A\alpha^n, -\infty < n < \infty$$

when A and α are **real** numbers, it's a real exponential sequence.

$\alpha=1.2$



$\alpha=0.9$



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* The *periodicity* of sinusoidal and exponential sequence *

- If $2\pi/\omega_0$ is an *rational number*, the sequence $x[n]=A\cos(\omega_0 n+\phi)$ is *periodic*.
- Otherwise, it's *aperiodic*.

Why?

How to determine the *fundamental period*?

To verify the above fact, consider

$$x_1[n] = A\cos(\omega_0 n + \phi)$$

$$x_2[n] = A\cos(\omega_0(n+N) + \phi)$$

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$$x_2[n] = \cos(\omega_0(n+N) + \phi)$$

$$= \cos(\omega_0 n + \phi) \cos \omega_0 N - \sin(\omega_0 n + \phi) \sin \omega_0 N$$

which will be equal to $\cos(\omega_0 n + \phi) = x_1[n]$

only if **$\sin \omega_0 N = 0$ and $\cos \omega_0 N = 1$** .

These two conditions are met if and only if

$$\omega_0 N = 2\pi r \quad (2.53a)$$

$$\text{or} \quad 2\pi / \omega_0 = N/r \quad (2.53b)$$

where N and r are positive integers.

Smallest value of N satisfying $\omega_0 N = 2\pi r$
is the *fundamental period* of the sequence.

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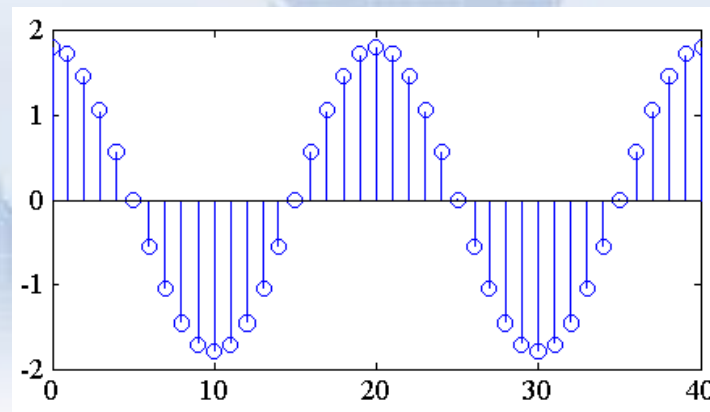


Example - $x[n]=\sin(3n)$ is an **aperiodic** sequence.

$y[n]=\cos(0.1\pi n)$ is a **periodic** sequence.

Hence $N = 2\pi r / \omega_0$
 $= 20$ for $r = 1$

$$\omega_0 = 0.1\pi$$



Read Example 2.11 and 2.12 by yourself.

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■ Two Properties of Sinusoidal Sequence

1. Consider: $x_1[n] = A \cos(\omega_1 n + \phi)$, $x_2[n] = A \cos(\omega_2 n + \phi)$

where, $0 \leq \omega_1 < 2\pi$, and $2\pi k \leq \omega_2 < 2\pi(k+1)$

k is an any positive integer.

$$\text{If} \quad \omega_2 = \omega_1 + 2\pi k \quad (2.55)$$

$$\text{Then } x_2[n] = A \cos((\omega_1 + 2\pi k) n + \phi) = x_1[n]$$

ω_2 is equivalent to ω_1 . (indistinguishable)

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i.e, ω_1 is equivalent to ω_2 modulo 2π .

$$\omega_2 = \omega_1 + 2\pi k$$

$$\{\text{range } 0 \leq \omega_1 < 2\pi\}$$

folding frequency

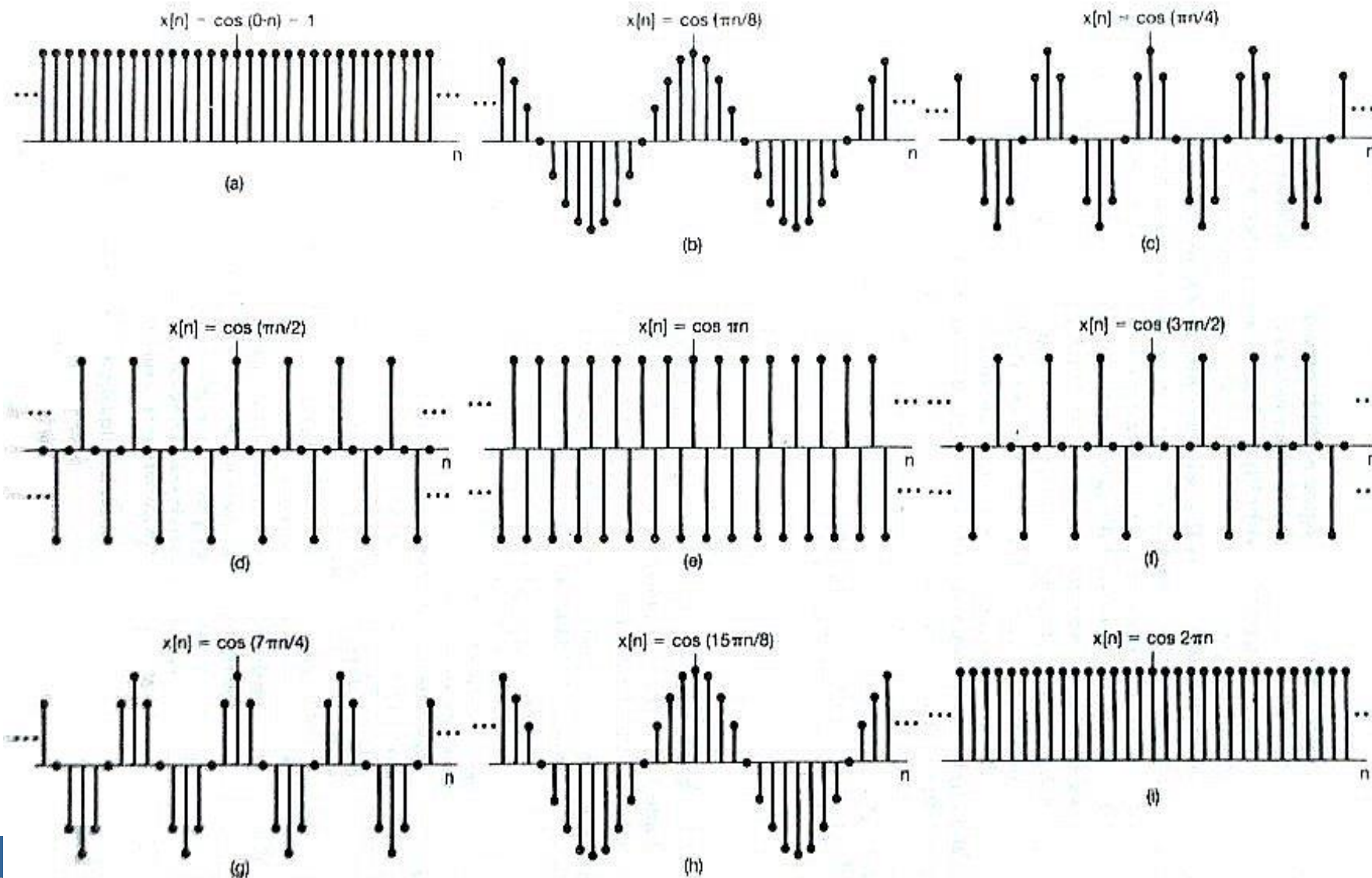
2. The highest ω of Sinusoidal Sequence is π .

Observe the Oscillation of **Sinusoidal Sequence**

$$\cos(1.1\pi n) = \cos((2\pi - 1.1\pi)n) = \cos(0.9\pi n)$$

$$\cos(0.8\pi n) = \cos((2\pi - 0.8\pi)n) = \cos(1.2\pi n)$$

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2.4.2 Sequence Generation Using MATLAB

Example -

$$x[n] = \exp\left(-\frac{1}{12} + j\frac{\pi}{6}\right)n$$

n=0:40;

K=1;

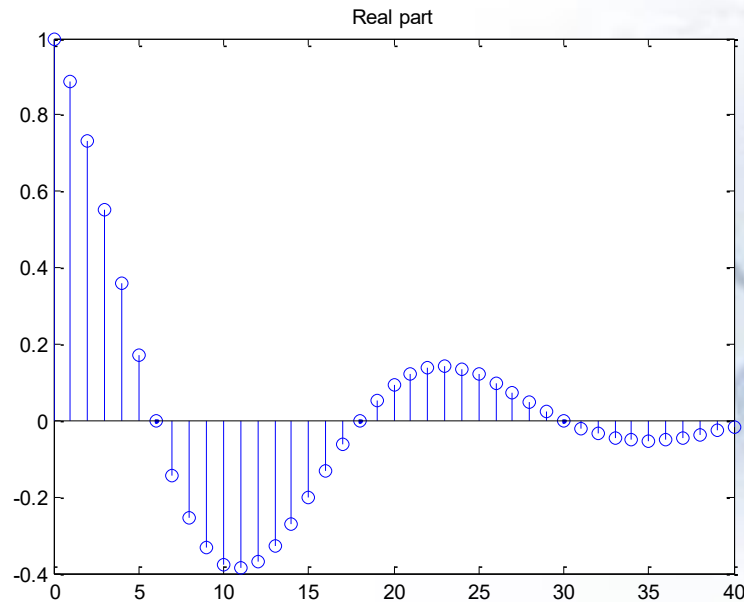
c=(-1+j*pi)/12;

x=K*exp(c*n);

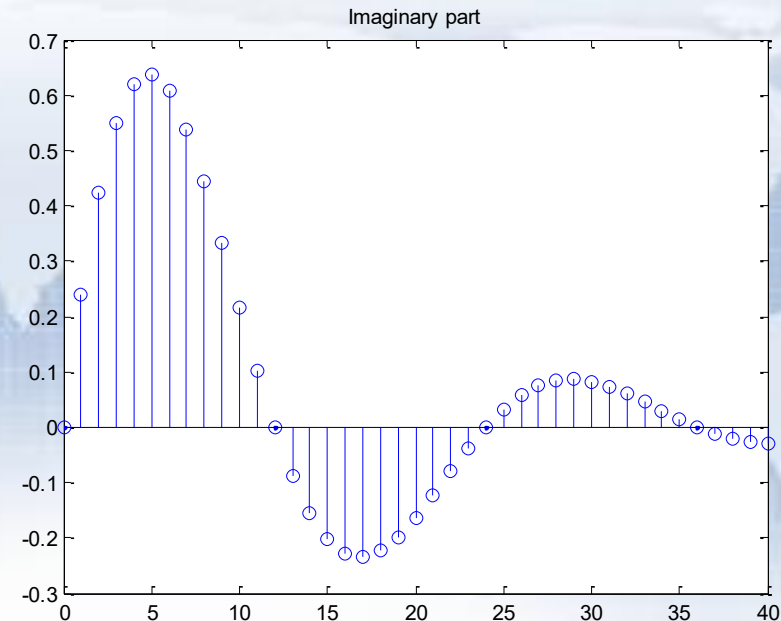
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```
stem(n, real(x));  
title('Real part');
```



```
stem(n, imag(x));  
title('Imaginary part');
```

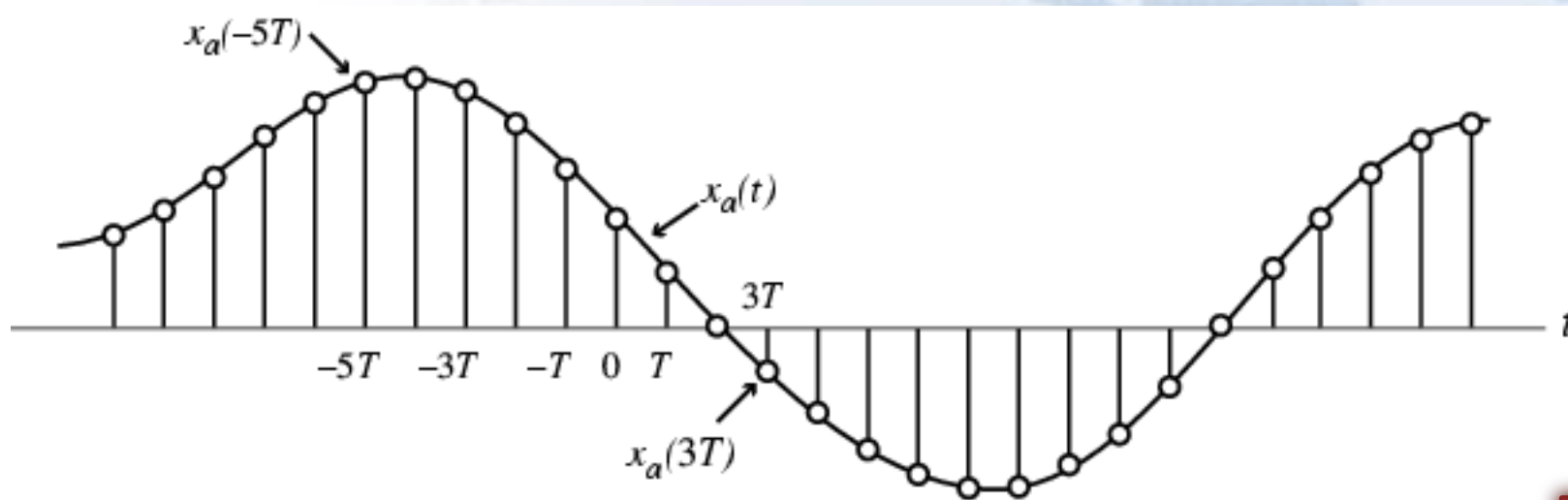


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§ 2.5 The Sampling Process

- Often, a discrete-time sequence $x[n]$ is developed by uniformly sampling a continuous-time signal $x_a(t)$ as indicated below



$$x[n] = x_a(t)|_{t=nT} = x_a(nT), \quad n = \dots, -2, -1, 0, 1, 2, \dots$$

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$$x[n] = x_a(t) \big|_{t=nT} = x_a(nT)$$

- Time variable t of $x_a(t)$ is related to the time variable n of $x[n]$ only at discrete-time instants t_n given by

$$t_n = nT = \frac{n}{F_T} = \frac{2n\pi}{\Omega_T}$$

Here: $F_T = 1/T$ ---sampling frequency

$\Omega_T = 2\pi F_T$ ---sampling angular frequency

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- Sampling a continuous-time *Sinusoidal* signal

$$x(t) = A \cos(2\pi f_o t + \phi) = A \cos(\Omega_o t + \phi)$$

The corresponding discrete-time signal is

$$\begin{aligned} x[n] &= A \cos(\Omega_o n T + \phi) = A \cos\left(\frac{2\pi\Omega_o}{\Omega_T} n + \phi\right) \\ &= A \cos(\omega_o n + \phi) \end{aligned}$$

where $\omega_o = 2\pi\Omega_o / \Omega_T = \Omega_o T$

is the **normalized digital angular frequency** of $x[n]$

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- **Frequency terms**

- f_0 , frequency, Hz, cycles/sec.
- $\Omega_0 = 2\pi f_0$, angular frequency, rad./sec.
- F_T , sampling frequency, Hz, sample/sec.
- $T = 1/F_T$, sampling interval, Sec., sec./sample
- $\omega_0 = \Omega_0 T$, digital angular frequency, rad./sample, sometimes, normalized digital angular frequency

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- **Note:** The **Ambiguity** in the DT Representation of CT Signals

Example 2.15 Three continuous-time signals
of frequencies 3 Hz, 7 Hz, and 13 Hz,

$$g_1(t) = \cos(6\pi t), g_2(t) = \cos(14\pi t), g_3(t) = \cos(26\pi t)$$

While sample distance $T=0.1\text{sec}$, the derived sequences are:

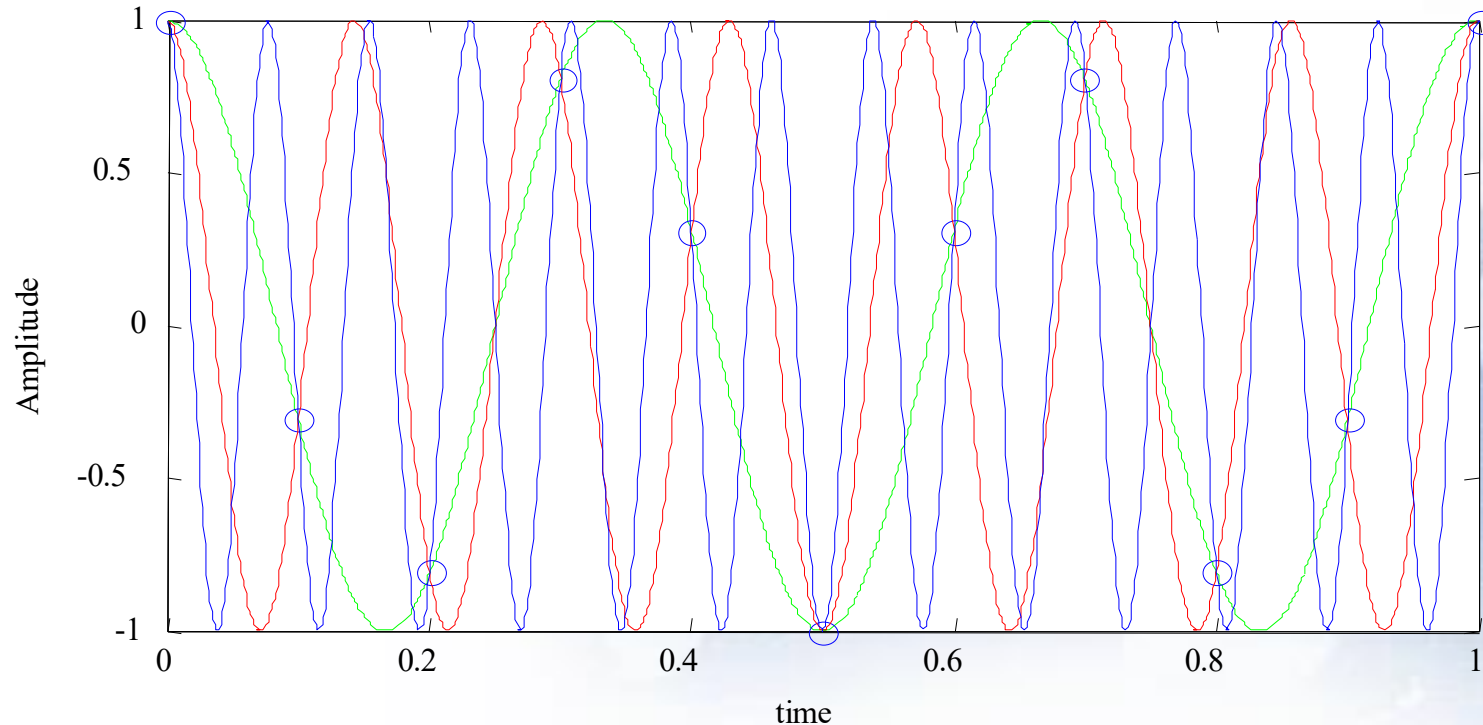
$$g_1[n] = \cos(0.6\pi n), g_2[n] = \cos(1.4\pi n), g_3[n] = \cos(2.6\pi n)$$

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- Plots of these sequences (shown with circles) and their parent time functions :



- Note that each sequence has exactly the same sample value for any given n**

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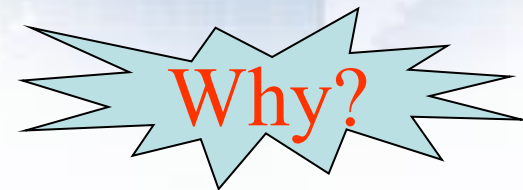


- This fact can also be verified by observing that

$$g_2[n] = \cos(1.4\pi n) = \cos((2\pi - 0.6\pi)n) = \cos(0.6\pi n)$$

$$g_3[n] = \cos(2.6\pi n) = \cos((2\pi + 0.6\pi)n) = \cos(0.6\pi n)$$

- As a result, all three sequences are **identical** and it is difficult to associate a unique continuous-time function with each of these sequences.



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- The above phenomenon of a continuous-time signal of higher frequency acquiring the identity of a sinusoidal sequence of lower frequency after sampling is called **aliasing**.

Recall

$$\omega_0 = 2\pi\Omega_0/\Omega_T$$

Thus if $\Omega_T > 2\Omega_0$, then ω_0 of the discrete-time signal obtained by sampling the parent continuous-time sinusoidal signal will be in the range $-\pi < \omega < \pi$.

→ **No aliasing**

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- On the other hand, if $\Omega_T < 2\Omega_0$, the normalized digital angular frequency will **fold over** into a lower digital frequency

$\omega_0 = (2\pi\Omega_0 / \Omega_T)_{2\pi}$ in the range $-\pi < \omega < \pi$ because of aliasing

- Hence, to prevent aliasing, the sampling frequency Ω_T should be **greater than 2 times** the frequency Ω_0 of the sinusoidal signal being sampled (*sampling theorem*, Section 3.8.1)

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Homework



- 2.3, 2.4(a) (b), 2.5, 2.9, 2.19, 2.21, 2.27, 2.28, 2.47, 2.51;
- M2.2, M2.3, M2.4